

ECS E 2500

Feb. 6th

Lec 8

Plan: Discrete Random Variable

□ Random variable — Discrete

In many engineering applications, we are interested in some numerical attributes of the outcome, e.g.,

of heads in 5 coin flips;

of packets transmission in a given time interval;

of failure/outage in a residential area

A random variable X is a function that assigns each outcome in the sample space $s \in S$ to a real number $X(s)$

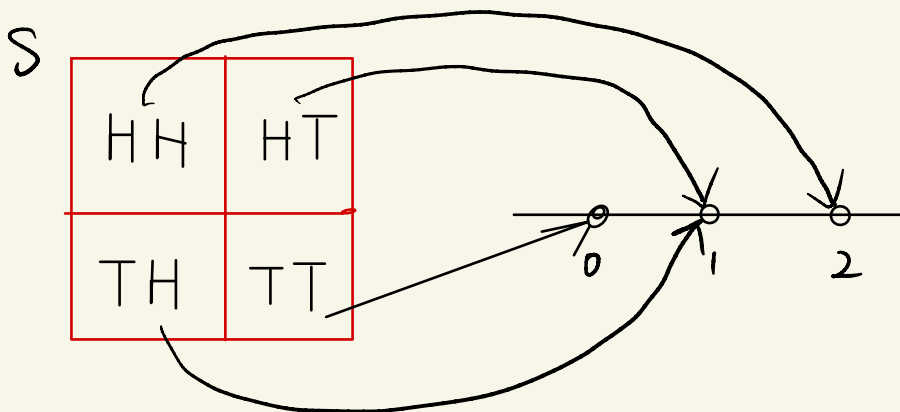
$$X: S \rightarrow \mathbb{R}$$

Ex: $S = \{HH, HT, TH, TT\}$

Introduce a random variable X

s	$X(s)$
HH	2
HT	1
TH	1
TT	0

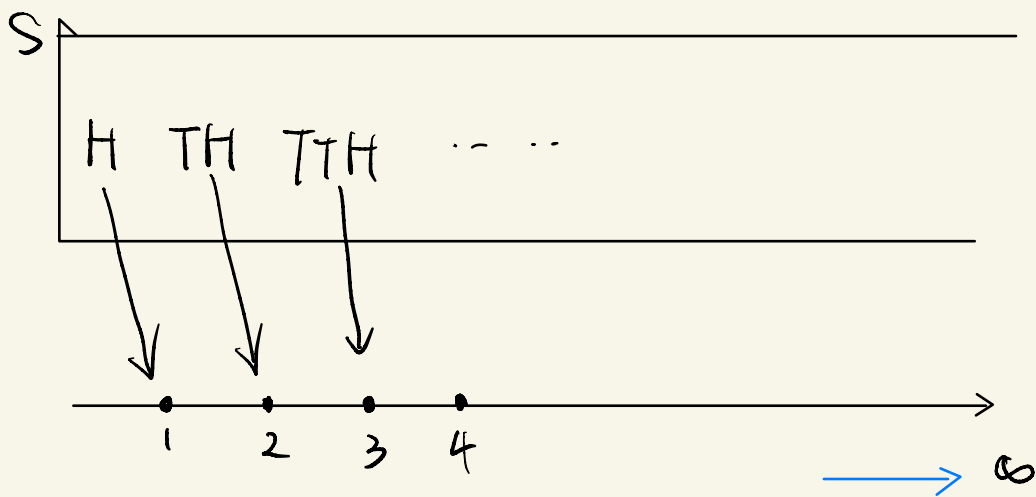
Def: A **discrete** random variable X is a function from S to a discrete set of values on the real space $\mathbb{R}$ / real line.



o We can have an infinite number of possible values.

Example: Flip a coin until you get a head.

Record the number of times we flipped before that.



Remember that we defined **probability** on **events**;
i.e., subsets of sample space S . But we
can see how the variable X inherits this concept.

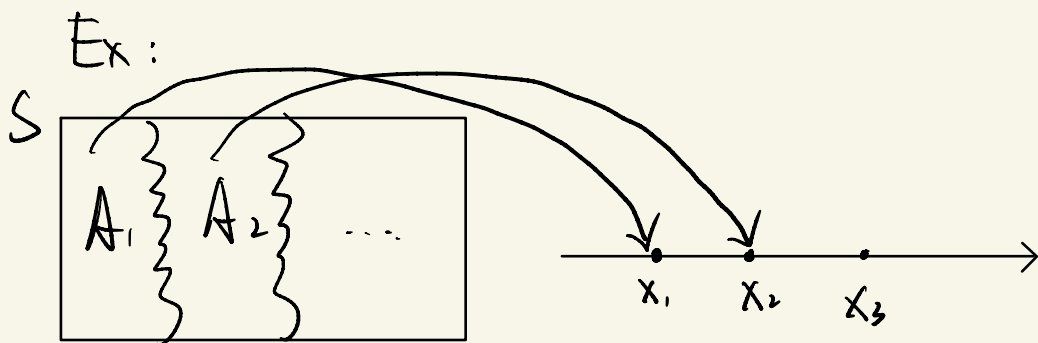
Example: Event $A := \{X \geq 3\}$ in flipping until a head
a set of real values

$$P(A) = P(\{s \mid X(s) \geq 3\})$$

$$= P(\{TTH, TTTH, TTTTH, \dots\})$$

a set of outcomes

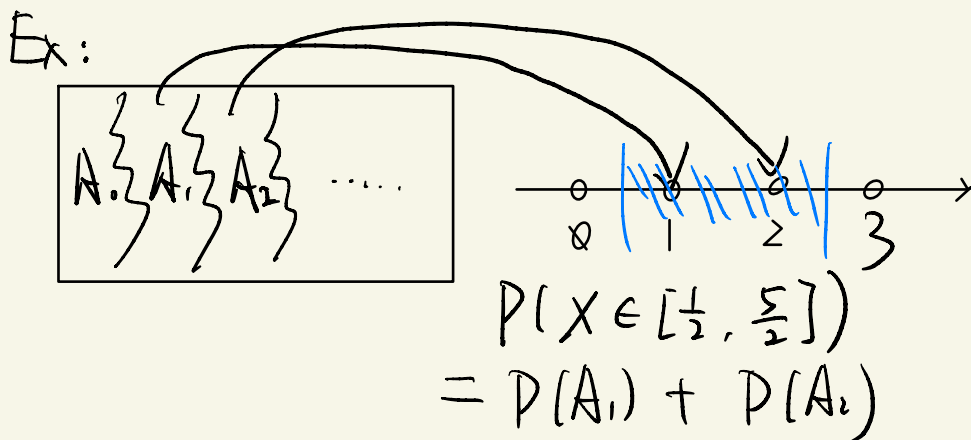
We can see that the random variable induces a partition of events on S .



$$A_i = \{s \in S \mid X(s) = x_i\}.$$

Then for any subset B of the real value, we can define probability $P(X \in B)$ as

$$P(X \in B) = P(\{s \in S \mid X(s) \in B\})$$



Therefore, from now on, we will exclusively talk about random variables since they assign a numerical value to each outcome of experiment and will allow us to talk about averages, variance in an unified way.

- The **key concept** of discrete random variable
 - Probability Mass Function (PMF)

The PMF of a discrete variable X is defined as

$$P_X(x) = P(X=x) = P(\{s \mid X(s)=x\}),$$

Big case $\uparrow$ Small case $\uparrow$
Random variable value/output of random variable

for any $x \in \mathbb{R}$

Since we focus on discrete random variable,

$P_X(x)$ is non-zero only at a set of values

$$S_X := \{x_1, x_2, x_3, \dots\}$$

We can think of S_X as the sample space of an experiment and we assign probability to events.

Some properties of PMF for discrete random variables:

1) $P_X(x) \geq 0$ for all x

2) $\sum_{x \in S_X} P_X(x) = 1$

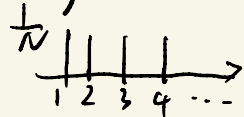
3) $P(X \in B) = \sum_{x \in B} P_X(x)$, for $B \subseteq S_X$

All three properties can be derived from three Axioms.

□ Seven important random variables with their corresponding PMFs

① Uniform random variable on $\{1, 2, 3, \dots, N\}$

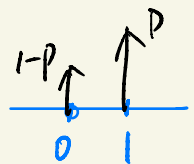
$$P_X(x) = \frac{1}{N}, \quad x \in \{1, 2, \dots, N\}$$



② Bernoulli random variable: maps successes and failures to 1's and 0's,

$$P_X(0) = 1 - p, \quad P_X(1) = p$$

$\uparrow$
failure $\uparrow$
success



- ③ Geometric random variable: Measure # of Bernoulli trials before a success.

$$P_X(1) = P$$

$$P_X(2) = (1-P)P \quad P_X(x) = (1-P)^{x-1}P,$$

$$P_X(3) = (1-P)^2P \quad \text{for all } x$$

...

- ④ Binomial random variable: Measure # of success in n Bernoulli trials.

$$P_X(k) = P(k \text{ successes in } n \text{ trials})$$

$$= C_k^n P^k (1-P)^{n-k},$$

$$k=0, 1, 2, \dots, n$$

Preview: If we don't know the PMF, we can estimate it by doing a large number of random experiments and count

$$\frac{\text{\# of time } X = x_k}{\text{\# of experiments}} \xrightarrow{n \rightarrow \infty} P_X(x_k)$$

⑤ Poisson Random Variable:

ex: # of cars passing in a crossroad

of phone calls in a given cell

of packets arriving in an interval

PMF of Poisson RV

$$P_X(k) = \frac{\alpha^k}{k!} e^{-\alpha}, \quad k=0,1,2,\dots$$

where α is the average number of events in a given interval.

Ex: X is the number of packets arriving in a given 10-minute window;

α is the average number of packets expected arriving in 10-min window.

Estimate
from historic
data

What is $P(\text{more than 4 packets arrived in 10 min})$ given that average $\alpha = 2$?

$$P(X \geq 4) = 1 - P(X \leq 3) = 1 - \sum_{k=0}^3 \frac{2^k}{k!} e^{-2}.$$